

CHILL DIVISION

Chill Division Automations Guide

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Cultivation Facility Automations Guide

v0.1.0 (DRAFT) "The little details matter"

Status: Draft skeleton. This is a new companion document to the Site Design Guidelines, Cultivation SOP, and Security Policies & Procedures. Its purpose is to give a reader the essential, practical detail needed to actually build the automations the other documents assume — what sensors are needed, what Home Assistant helpers to create, and when to run logic in ESPHome versus Home Assistant. Section headings below are placeholders; `TODO` marks sections not yet written.

Introduction

TODO — scope, audience, and how this guide relates to the other three documents.

This guide assumes a locally-controlled Home Assistant setup with ESPHome microcontrollers, consistent with the Site Design Guidelines. See the Chill Division M5Stack ESPHome repository for per-device YAML: <https://github.com/Chill-Division/M5Stack-ESPHome>.

All terminology follows RFC 2119 (MUST / SHOULD / MAY).

1. Architecture overview

TODO — local-first control, ESPHome vs Home Assistant responsibilities, PoE/wired-first principle, network dependencies.

2. ESPHome vs Home Assistant: where should logic live?

TODO — decision guidance: safety-critical / fast-loop / must-survive-HA-restart logic in ESPHome; orchestration, scheduling, cross-device logic in Home Assistant. Worked examples.

3. Sensors and what they are for

TODO — SCD4x (CO2/VPD), ENV.IV (temp/humidity/pressure), PM2.5, ENV.Pro (VOC/IAQ), substrate sensors (THC-S/Teros12), MQ2 smoke, ultrasonic reservoir level. Placement, cabling, i2c limits.

4. Home Assistant helpers required

TODO — the input_number / input_boolean / template / schedule / threshold helpers each automation depends on. One subsection per subsystem.

5. Environmental control automations

TODO — VPD calculation, CO2 injection setpoint/solenoid control, humidification/dehumidification coordination, air conditioning, ventilation.

6. Lighting and photoperiod automations

TODO — photoperiod start/end, dimming ramps, power-outage-safe "start off" behaviour, room lighting alerts.

7. Irrigation / fertigation automations

TODO — batch tank refills, dosing, solenoid sequencing, crop-steering phases (P0–P3), dryback targets.

8. Safety, alerting, and failsafes

TODO — smoke/CO2 alerts, duress button flow, low-battery and stuck-sensor alerts, power-failure notifications, fallback behaviour.

9. Example configurations

TODO — copy-pasteable ESPHome snippets and Home Assistant automation/template examples, linking to the M5Stack ESPHome repo.

Changelog

- v0.1.0 — 2026-07-03
 - Created draft skeleton and proposed section outline.